

MOHAMED AMINE HERCH

Industrial Electronics Engineer | Mobile Robotics & Embedded Systems
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PROFIL

Industrial Electronics Engineer specializing in mobile robotics, industrial automation, and embedded systems. Hands-on experience integrating autonomous mobile robots and developing robotic systems using ROS 2 and Open-RMF, with real-world experience in navigation, localization, robot communication, and system troubleshooting. Skilled in C/C++, Python, industrial communication, sensors and actuators, and real-time robotic systems.

EDUCATION

National Engineering School of Sousse (Tunisia)

Engineering Degree in Industrial Electronics | 2023 – 2026

Preparatory Institute for Engineering Studies (Monastir , Tunisia)

Scientific Preparatory Cycle | 2022 – 2023

Preparatory Institute for Engineering Studies (Sfax , Tunisia)

Scientific Preparatory Cycle | 2021 – 2022

TECHNICAL SKILLS

- Robotics & Fleet Management:** ROS 2 Jazzy, Nav2, SLAM, Open-RMF, AMR/AGV Integration, Autonomous Navigation
- Digital Twin & 3D:** Unity 6.3 LTS, Real-Time 3D Visualization, Digital Twin, ROS-TCP-Connector, 3D Scanning
- Mechatronics & Automation:** Sensors & Actuators, Electromechanical Systems, Motors & Motor Control, Industrial Automation, Measurement ,
- Embedded Systems & IoT:** STM32, ESP32, ESP8266, Arduino, FPGA/VHDL, RFID, MQTT, Serial Communication
- Programming & Tools:** C/C++, Python/relpy, MATLAB, Node-RED, Git, Linux, Proteus, Altium, Xilinx ISE
- Prototyping & Fabrication:** 3D Printing, Laser Engraving

LANGUAGES

Arabic: Native
French: Intermediate
English: Intermediate

LEADERSHIP & ACTIVITIES

President : ENISO Team | 2024 – 2025

- Led a multidisciplinary team of engineering students.
- Organized technical training sessions covering SolidWorks, MATLAB, STM32, Arduino, and ESP32.
- Coordinated major student events, including ESC V9.

PROFESSIONAL EXPERIENCE

Final-Year Project (Heterogeneous AMR Fleet Integration and Digital Twin Supervision) :CNIMI Laboratory | drummondville, Canada, 2026

- Developed and configured a heterogeneous AMR fleet management system using ROS 2 Jazzy and Open-RMF, enabling multi-robot coordination across different manufacturers.
- Developed a Unity-based Digital Twin for real-time 3D visualization, robot state monitoring, and fleet supervision through ROS-TCP-Connector.
- Integrated MiR250 + UR10e, MiR600, and Omron LD-90 into the Open-RMF framework, including maps, navigation routes, traffic rules, and fleet coordination.
- Keywords:** Open-RMF, ROS 2 Jazzy, Digital Twin, Unity, AMR, Multi-Robot Coordination, ROS-TCP-Connector, ARCL, Python, C++, Linux

Intern : Rosenberger | Sousse, Tunisia, 2025

- Developed autonomous navigation and mapping simulations using ROS and SLAM algorithms.
- Designed and built a physical robotic prototype integrating a Raspberry Pi, LiDAR, and DC motors with encoders.
- Keywords:** Raspberry Pi, ROS, LiDAR, SLAM, DC Motors with Encoders, Autonomous Navigation, Robotics

Intern : Leoni | Sousse, Tunisia, 2024

- Observed all stages of wiring harness production.
- Studied assembly processes and quality control procedures.

Maker :Fablab ENISO | 2024 – Présent

- Senior Maker with experience mentoring and supervising robotics and mechatronics projects.
- Supported projects from concept and design through prototyping, integration, and testing.
- Delivered technical training sessions and hands-on workshops in robotics, electronics, and digital fabrication.
- Hands-on experience with 3D printing, laser engraving, and 3D scanning.
- Keywords:** Robotics, Mechatronics, Project Supervision, Technical Training, Prototyping, Embedded Systems, 3D Printing, Laser Engraving, 3D Scanning .

ACADEMIC PROJECTS

SmartTime IoT : Connected Employee Attendance System

- Developed an ESP8266-based RFID attendance system with MQTT/HiveMQ communication and Node-RED real-time monitoring.
- Keywords:** ESP8266, RFID, MQTT, HiveMQ, Node-RED, IoT

Sign Language Recognition

- Developed a gesture recognition system using MATLAB and image processing techniques to identify sign language gestures.
- Explored machine learning and computer vision methods for gesture classification.
- Keywords:** MATLAB, Image Processing, Machine Learning, Computer Vision

Line-Following Robot

- Developed an autonomous line-following robot using Pololu encoders and sensor fusion for accurate trajectory tracking.
- Keywords:** Pololu Encoders, Sensor Fusion, PID Control, Embedded Systems, Robotics